

Chapter 6: Flexible Parametric Models

Modelling Survival Data in Medical Research

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Bridging Cox and parametric models

1. Why flexible models?
2. The piecewise exponential model

Spline functions

3. Modelling with splines
4. B-splines
5. Restricted cubic splines
6. Number and position of knots

Flexible regression models

7. Flexible models for the hazard
8. Flexible models for the log-cumulative hazard
9. The Royston–Parmar model
10. Flexible proportional odds models

Goal: smooth, parametric estimates of the baseline hazard that the Cox model cannot provide.

Why Flexible Models?

Standard parametric models are too rigid:

The distributions of Chapter 5 give baseline hazards governed by just a few parameters, so they have **limited ability** to capture the true shape of a baseline hazard.

The Cox model's flexibility is often overplayed:

- ▶ The estimated baseline hazard is a **step function** jumping at each event time, it can behave erratically
- ▶ The estimated survivor function is constant between events, so survival at a precise time may not be estimable
- ▶ Survival beyond the longest event time cannot be estimated at all

Flexible parametric models

Bridge the gap between the Cox semi-parametric model and fully parametric models.

What they offer:

- ▶ A **smooth, parametric** baseline hazard
- ▶ Survival estimates at **any** time, including extrapolation
- ▶ Often a better summary than a step function

Especially valuable when standard parametric models do not adequately depict the underlying baseline hazard.

The Piecewise Exponential Model

The idea:

Model the baseline hazard explicitly with **one parameter per interval** between event times. Within each interval the hazard is constant. With r ordered event times

$t_{(1)} < t_{(2)} < \dots < t_{(r)}$, set

$$h_0(t) = \lambda_j, \quad t_{(j-1)} \leq t < t_{(j)}$$

with $\lambda_j > 0$, and $t_{(0)} = 0$. To keep $\lambda_j > 0$, often parameterise $\lambda_j = \exp(\phi_j)$ and treat the ϕ_j as unknown.

Why “piecewise exponential”?

A constant hazard in an interval $\Rightarrow$ exponentially distributed times in that interval. The model fits a separate exponential in each piece. **Knots (cut points)**:

The points $t_{(1)}, \dots, t_{(r-1)}$ where the hazard changes are called the **cut points** or **knots**.

The full model

For individual i with covariates x_i :

$$h_i(t) = \exp(\beta' x_i) h_0(t)$$

— a proportional hazards model with an explicit step-function baseline.

Piecewise Exponential: Cumulative Hazard and Fitting

Cumulative baseline hazard:

Integrating the step-function hazard gives a piecewise-*linear* cumulative hazard:

$$H_0(t) = \sum_{j=1}^r \lambda_j \Delta_j(t)$$

where

$$\Delta_j(t) = \begin{cases} 0 & t < t_{(j-1)} \\ t - t_{(j-1)} & t_{(j-1)} \leq t < t_{(j)} \\ t_{(j)} - t_{(j-1)} & t \geq t_{(j)} \end{cases}$$

Then $H_i(t) = \exp(\beta' x_i) H_0(t)$ and $S_i(t) = \exp\{-H_i(t)\}$.

Fitting by maximum likelihood:

Maximise the full parametric likelihood

$$L = \prod_{i=1}^n \{h_i(t_i)\}^{\delta_i} S_i(t_i)$$

over the λ_j and the β 's.

Computational challenges

With one parameter per inter-event time, the log-likelihood is hard to maximise, and the usual information-matrix standard errors are unstable.

Bootstrapping for standard errors: resample n observations with replacement, re-estimate, repeat 500+ times; the SD of the bootstrap estimates gives the standard error.

Piecewise Exponential: Choosing Intervals

One parameter per event time is too many:

It is computationally demanding and does *not* give an effective summary of the hazard; the model is grossly over-parameterised. **Use**

fewer, well-placed intervals:

- ▶ Usually no more than **10 intervals** are needed
- ▶ Choose end points so each interval contains a **similar number of events**
- ▶ Or make intervals narrow where the hazard changes quickly, wider where it changes slowly

An informed choice of boundaries should approximate any baseline hazard reasonably well.

Comparing models: the AIC

$$\text{AIC} = -2 \log \hat{L} + 2q$$

where q is the total number of parameters.
Smaller AIC is preferred.

Caution on comparisons

The Cox model's AIC comes from a *partial* likelihood, so it **cannot** be compared with the *full*-likelihood AIC of piecewise exponential models.

Example: Hypernephroma

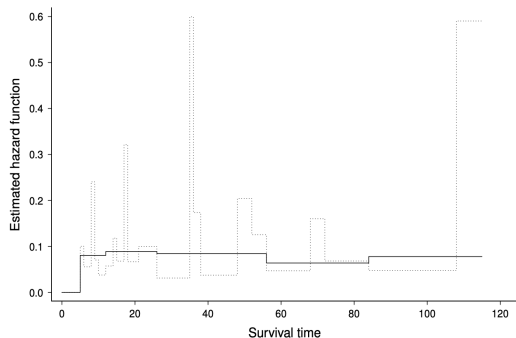


Figure: Piecewise exponential baseline hazards: dotted (event-time) and solid (5 intervals).

Two piecewise models compared:

Model	Params	AIC
22 intervals	25	298.70
5 intervals	8	283.48
exponential	4	298.76

Findings:

- ▶ Hazard ratios are stable across models (nephrectomy ≈ 0.24 ; over-70 $\approx 4-5$)
- ▶ The model with 22 intervals is grossly over-parameterised
- ▶ The 5-interval model (knots at quintiles: 12, 26, 56, 84 months) has the **lowest AIC**
- ▶ It also beats a constant-hazard exponential model

What is a spline?

Divide the time scale into intervals and fit a **polynomial** of t within each interval, constraining the pieces to **join smoothly** at the interval boundaries.

Key terminology:

- ▶ **Knots:** the interval boundaries
- ▶ **Knot sequence:** the full set of knots
- ▶ **Boundary knots:** at the extreme values of t
- ▶ **Internal knots:** between the boundaries
- ▶ **Degree:** the degree of the constituent polynomials

Why cubic splines (degree 3)?

They strike a compromise between simplicity and flexibility, and are:

- ▶ Straightforward to compute
- ▶ Numerically stable
- ▶ Smooth

More knots $\Rightarrow$ more flexibility:

A single internal knot gives a simple two-piece curve; adding internal knots produces increasingly complex shapes.

A linear combination of **basis functions** forms the spline. The coefficients are estimated when fitting the model.

Definition:

A B-spline of degree k with m internal knots is built from $m + k + 1$ basis functions, each of degree k :

$$B(t) = \sum_{j=1}^{m+k+1} \alpha_j B_{j,k}(t), \quad t_{\min} \leq t \leq t_{\max}$$

Cubic B-spline ($k = 3$): $m + 4$ basis functions

$$B(t) = \sum_{j=1}^{m+4} \alpha_j B_{j,3}(t)$$

The $m + 4$ coefficients α_j are estimated from the data.

Augmenting the knot sequence:

To define the basis functions recursively, repeat the boundary knots three times at each end:

$$\underbrace{t_{\min}, \dots, t_{\min}}_4, t_1, \dots, t_m, \underbrace{t_{\max}, \dots, t_{\max}}_4$$

giving $m + 8$ augmented knots $\tau_1, \dots, \tau_{m+8}$.

Each basis function is local:

A B-spline basis function is non-zero only over a limited span of t , which gives the method its numerical stability.

B-splines: The Recurrence Relationship

Start with degree zero (indicator functions):

$$B_{j,0}(t) = \begin{cases} 1 & \tau_j \leq t < \tau_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

Build higher degrees recursively:

$$B_{j,k+1}(t) = \omega_{j,k+1}(t) B_{j,k}(t) + \{1 - \omega_{j+1,k+1}(t)\} B_{j+1,k}(t)$$

where

$$\omega_{j,k}(t) = \begin{cases} \frac{t - \tau_j}{\tau_{j+k} - \tau_j} & \tau_{j+k} \neq \tau_j \\ 0 & \text{otherwise} \end{cases}$$

The procedure:

- 1 Start with the degree-0 indicator functions
- 2 Apply the recurrence with $k = 0$ to get degree-1 functions
- 3 Apply again with $k = 1$ for degree 2
- 4 Apply again with $k = 2$ for degree 3

This yields the $m + 4$ cubic basis functions $B_{j,3}(t)$.

The recurrence is the engine of the B-spline:
each basis function is built from two
lower-degree functions, weighted by their
position relative to the knots.

B-spline: A Worked Illustration

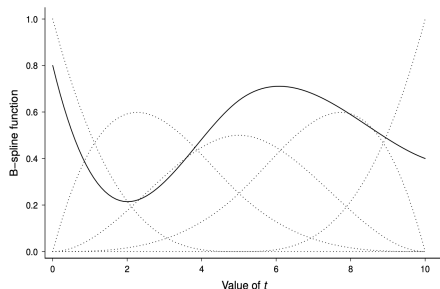


Figure: Five cubic B-spline basis functions (dotted) with one internal knot at $t = 5$, and their linear combination forming a fitted spline (solid).

Setup: survival times 0 to 10, one internal knot at $t = 5$. Augmented knots: $(0, 0, 0, 0, 5, 10, 10, 10, 10)$.

Result: five cubic basis functions $B_{1,3}, \dots, B_{5,3}$, each a cubic in t with smooth joins at $t = 5$.

Building a fitted curve:

With coefficients, say, $(0.8, -0.3, 1.2, 0.5, 0.4)$:

$$B(t) = \sum_{j=1}^5 \alpha_j B_{j,3}(t)$$

gives a smooth curve that dips, peaks near $t = 6$, then declines slowly.

Different coefficients give different shapes — splines can represent a wide variety of functional forms, ideal for a baseline hazard or its logarithm.

Restricted (Natural) Cubic Splines

Problem with standard splines:

Polynomial splines can behave erratically near boundary knots and are not well-behaved for extrapolation.

Key idea:

Force the function to be **linear in the tails**: below $k_{\min}$ and above $k_{\max}$.

Benefit:

More stable extrapolation and fewer basis functions than standard B-splines.

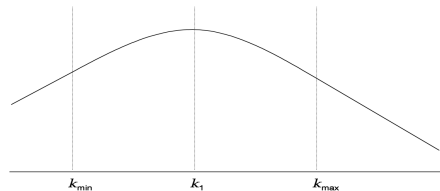


Figure: Restricted cubic spline with one internal knot at k_1 , and boundary knots at $k_{\min}$ and $k_{\max}$; linear beyond boundaries.

Restricted (Natural) Cubic Splines: Model Form

Model with m internal knots:

$$\gamma_0 + \gamma_1 t + \sum_{j=1}^m \gamma_{j+1} \nu_j(t)$$

Key properties:

- ▶ Total of $m + 2$ parameters
- ▶ Cubic spline between knots
- ▶ Linear beyond boundary knots

Special case:

$$m = 0 \Rightarrow \gamma_0 + \gamma_1 t$$

(reduces to simple linear regression)

Restricted Cubic Spline: Basis Function

The basis function for the j th internal knot k_j :

$$\nu_j(t) = (t - k_j)_+^3 - \lambda_j(t - k_{\min})_+^3 - (1 - \lambda_j)(t - k_{\max})_+^3$$

where

$$\lambda_j = \frac{k_{\max} - k_j}{k_{\max} - k_{\min}}$$

and

$$(t - a)_+^3 = \max\{0, (t - a)^3\},$$

(zero if $t < a$).

Worked example: one knot at $k_1 = 5$, boundaries 0 and 10, so $\lambda_1 = 0.5$.

$$\nu_1(t) = \begin{cases} -0.5 t^3 & 0 \leq t < 5 \\ (t - 5)^3 - 0.5 t^3 & 5 \leq t < 10 \end{cases}$$

For $t > 10$, this reduces to $\nu_1(t) = 375 - 75t$ — **linear**, as required.

Unlike the B-spline, the restricted cubic spline is defined for **all** values of t . So it can be used to extrapolate.

Recommended placement:

- ▶ **Boundary knots** at the smallest and largest observed event times
- ▶ **Internal knots** at evenly spaced percentiles of the uncensored event times:

Internal knots	Percentiles
1	50
2	33, 67
3	25, 50, 75

Usually no more than 4 or 5 knots are needed.

Choosing the number of knots:

- ▶ Start with boundary knots only, then add internal knots one at a time
- ▶ As knots move when their number changes, models are **not nested**
- ▶ So the $-2 \log \hat{L}$ likelihood ratio test **cannot** be used.

Use AIC or BIC

$$\text{AIC} = -2 \log \hat{L} + 2q$$

With p covariates and m internal knots in a restricted cubic spline, $q = p + m + 2$.

Increase the number of knots until the AIC (or BIC) stops decreasing.

Flexible Models for the Hazard Function

A spline-based proportional hazards model:

Start from $h_i(t) = \exp(\eta_i)h_0(t)$ with $\eta_i = \beta'x_i$.

Model the log-baseline hazard with a spline.

To guarantee $h_0(t) > 0$, build the model on the **logarithm** of the baseline hazard. Using a cubic B-spline with m internal knots:

$$\log h_0(t) = \sum_{j=1}^{m+4} \alpha_j B_{j,3}(t)$$

so the full model is

$$\log h_i(t) = \eta_i + \sum_{j=1}^{m+4} \alpha_j B_{j,3}(t)$$

Fitting:

Maximise the likelihood $\prod \{h_i(t_i)\}^{\delta_i} S_i(t_i)$. The survivor function needs the cumulative baseline hazard $H_0(t)$, obtained by **numerical integration** (no closed form on the log-hazard scale):

$$S_i(t) = \exp\{-e^{\eta_i} H_0(t)\}$$

Model building:

Fit models with increasing numbers of knots; choose the one with the smallest AIC or BIC.

Building on $\log h_0(t)$ keeps the fitted hazard non-negative for any spline coefficients.

Special Case: Recovering the Piecewise Exponential

Use degree-zero B-splines ($k = 0$):

The $m + 1$ basis functions $B_{j,0}(t)$ are indicator functions, unity in their interval, zero elsewhere. The baseline hazard becomes

$$h_0(t) = \exp \left\{ \sum_{j=1}^{m+1} \alpha_j B_{j,0}(t) \right\}$$

so $h_0(t) = \exp(\alpha_j)$ in the j th interval, **constant within each interval**.

This is the piecewise exponential model again!

- ▶ A degree-0 B-spline log-hazard model **is** a piecewise exponential model
- ▶ And it corresponds to a **Cox model** when the knots are placed at the event times

The piecewise exponential, the Cox model, and the flexible spline model are all members of one family, differing in how the baseline hazard is represented.

Example: Hypernephroma with a B-spline Hazard

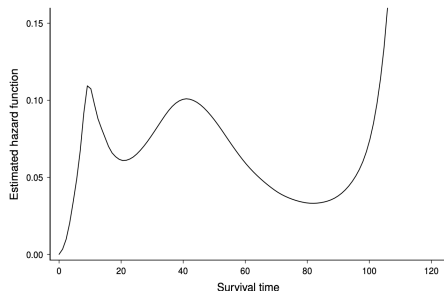


Figure: Baseline hazard from a cubic B-spline model with two internal knots fitted to the hypernephroma data; peaks near 10 and 40 days, then declines.

Model selection: the AIC is smallest with **two internal knots**, giving six basis functions:

$$\hat{h}_0(t) = \exp\{\hat{\alpha}_1 B_{1,3}(t) + \cdots + \hat{\alpha}_6 B_{6,3}(t)\}$$

Findings:

- ▶ The smooth baseline reveals two peaks (around 10 and 40 days)
- ▶ Hazard ratios (nephrectomy 0.25, over-70 3.97) match the piecewise model closely
- ▶ The precise baseline shape has **little impact on hazard ratios**
- ▶ But $AIC = 292.83$ here exceeds the piecewise model's — so the B-spline hazard does not improve the fit in this case

Why model the log-cumulative hazard?

Spline models for the hazard risk erratic behaviour near the shortest and longest event times. Models for the **log-cumulative hazard** are **less susceptible** to this.

The Weibull as the starting point.

For a Weibull model, the cumulative hazard is $H_i(t) = \exp(\eta_i)\lambda t^\gamma$, so

$$\log H_i(t) = \eta_i + \log \lambda + \gamma \log t$$

Writing $\gamma_0 = \log \lambda$, $\gamma_1 = \gamma$, $y = \log t$:

$$\log H_i(t) = \eta_i + \gamma_0 + \gamma_1 y$$

The Weibull log-cumulative hazard is **linear in** $y = \log t$.

This is the key insight: a Weibull model is just a **straight line** on the log-cumulative-hazard vs. log-time scale. Replace that line with a spline, and we get a flexible model.

The Royston–Parmar Model

Replace the straight line with a restricted cubic spline.

With m internal knots, in terms of $y = \log t$:

$$\log H_0(t) = \gamma_0 + \gamma_1 y + \sum_{j=1}^m \gamma_{j+1} \nu_j(y)$$

where each $\nu_j(y)$ is the restricted cubic spline basis function. For individual i :

$$\log H_i(t) = \eta_i + \gamma_0 + \gamma_1 y + \sum_{j=1}^m \gamma_{j+1} \nu_j(y)$$

Proposed by **Royston and Parmar (2002)**.

What this model is:

- ▶ Still a **proportional hazards** model in the covariates
- ▶ But survival times no longer follow a Weibull or *any* named distribution
- ▶ With **no internal knots**, it reduces exactly to the Weibull model

The basis function $\nu_j(y)$:

$$\begin{aligned} \nu_j(y) = & (y - k_j)_+^3 - \lambda_j (y - k_{\min})_+^3 \\ & - (1 - \lambda_j)(y - k_{\max})_+^3, \end{aligned}$$

cubic between the boundary knots, linear beyond them.

Survivor function (easily obtained):

Using $S_i(t) = \exp\{-H_i(t)\}$:

$$S_i(t) = \exp\left\{-\exp\left[\eta_i + \gamma_0 + \gamma_1 y + \sum_j \gamma_{j+1} \nu_j(y)\right]\right\}$$

Equivalently $S_i(t) = \{S_0(t)\}^{\exp(\eta_i)}$.

A major advantage of working on the log-cumulative-hazard scale: the survivor function follows directly, with no numerical integration.

Hazard function (by differentiation):

Since $h_0(t) = H_0(t) \frac{d \log H_0(t)}{dt}$, and differentiating the spline gives

$$h_i(t) = e^{\eta_i} \left\{ \gamma_1 + \sum_{j=1}^m \gamma_{j+1} \nu'_j(y) \right\} t^{-1} H_0(t)$$

where $\nu'_j(y)$ is the derivative of the basis function with respect to $y = \log t$.

Result: a fully parametric PH model whose baseline hazard is a **smooth function** of t , with $m + 2$ spline parameters $\gamma_0, \dots, \gamma_{m+1}$.

Fitting by maximum likelihood:

Maximise $\prod \{h_i(t_i)\}^{\delta_i} S_i(t_i)$ with S_i and h_i as just derived. Standard optimisation yields:

- ▶ Smooth fitted survivor and hazard functions
- ▶ Standard errors for all parameters
- ▶ Estimates of S and h at **any** time

A convenient practical fact:

The number of knots needed is **barely affected by covariate adjustment**. So:

- ① Select explanatory variables using a standard Weibull or Cox model
- ② Then fit Royston–Parmar models with increasing numbers of knots
- ③ Choose the model with the smallest AIC or BIC

Variable selection and knot selection can be done **separately** — a major practical simplification.

Example: Recurrence-Free Survival in Breast Cancer

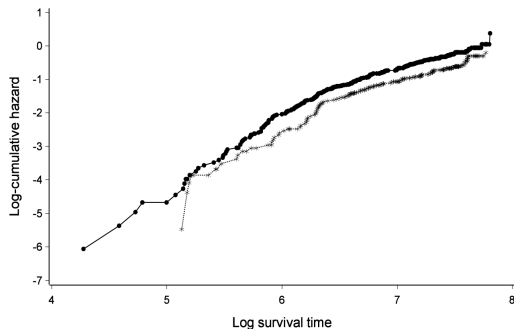


Figure: Log-cumulative hazard for non-tamoxifen (●) and tamoxifen (★) groups.

The study: German Breast Cancer Study Group; 686 patients with node-positive breast cancer; outcome is recurrence-free survival; the factor of interest is tamoxifen (vs. none), adjusting for age, menopausal status, tumour size/grade, nodes, and receptor status.

Reading the plot:

- ▶ The two curves are **not straight** $\Rightarrow$ a Weibull distribution is **not** appropriate
- ▶ Their vertical separation is roughly **constant** $\Rightarrow$ proportional hazards for treatment is plausible

Non-Weibull shape but proportional hazards:
an ideal case for the Royston–Parmar model.

Breast Cancer: Selecting the Number of Knots

AIC by number of internal knots:

Knots	AIC
0 (Weibull)	5182.4
1	5147.9
2	5147.1
3	5146.1
4	5148.2

Findings:

- ▶ Large improvement from 0 to 1 knot over Weibull
- ▶ Small gains for 2 and 3 knots
- ▶ AIC increases at 4 knots
- ▶ Best model: 3 knots (slightly better than 1 knot)

Breast Cancer: Risk-Adjusted Survival Functions

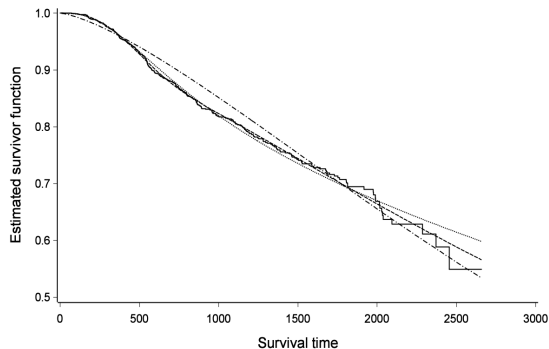


Figure: Risk-adjusted survivor functions: Cox (—), Weibull ($\cdots$), Royston–Parmar (1 knot: $\cdots$, 3 knots: $- - -$).

Weibull deviates from Cox baseline; One-knot model closely tracks Cox; Three knots improve fit in the tail.

Breast Cancer: Estimates (Royston–Parmar vs Cox)

Selected coefficient estimates:

Parameter	R–P estimate	Cox estimate
β_1 (Treat)	−0.339	−0.337
β_4 (Size)	0.0077	0.0077
β_5 (Grade)	0.282	0.280
β_6 (Nodes)	0.0497	0.0499

Key findings:

- ▶ Coefficients and standard errors are nearly identical to Cox
- ▶ Tamoxifen hazard ratio: $e^{-0.339} = 0.71$
- ▶ Consistent evidence of reduced risk under treatment

Breast Cancer: Fitted Baseline Hazard

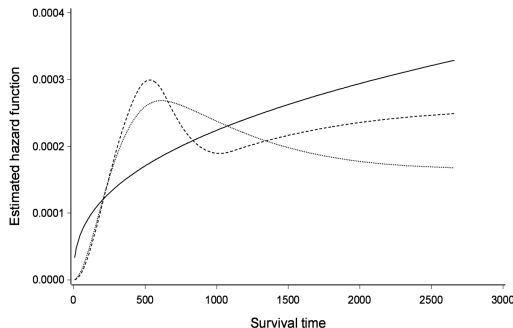


Figure: Adjusted baseline hazard functions for a fitted Weibull model (—), and Royston–Parmar models with one (···) and three knots (— —).

The Royston–Parmar model provides a smooth baseline hazard, revealing a clear unimodal shape and explaining the poor fit of the monotone Weibull model.

Flexible Proportional Odds Models

The general proportional odds model:

The odds of survival beyond t are

$$\frac{S_i(t)}{1 - S_i(t)} = \exp(\beta' x_i) \frac{S_0(t)}{1 - S_0(t)}$$

Starting from a log-logistic baseline,

$$\log \frac{S_0(t)}{1 - S_0(t)} = -\theta - \kappa \log t = \gamma_0 + \gamma_1 y$$

which is **linear in** $y = \log t$ (with $\gamma_0 = -\theta$, $\gamma_1 = -\kappa$).

Replace the line with a restricted cubic spline:

$$\log \frac{S_0(t)}{1 - S_0(t)} = \gamma_0 + \gamma_1 y + \sum_{j=1}^m \gamma_{j+1} \nu_j(y)$$

so the full flexible proportional odds model is

$$\log \frac{S_i(t)}{1 - S_i(t)} = \eta_i + \gamma_0 + \gamma_1 y + \sum_{j=1}^m \gamma_{j+1} \nu_j(y)$$

Exactly parallel to the Royston–Parmar hazard model, but built on the **log-odds** scale instead of the log-cumulative-hazard scale. Particularly suited to **unimodal** hazards.

Example: Breast Cancer Proportional Odds Model

Why try proportional odds here?

The unimodal hazard found earlier suggests a log-logistic-type model may fit the recurrence-free survival data better.

AIC by number of internal knots:

Knots	AIC
0	5154.0
1	5135.4
2	5135.8
3	5134.1
4	5136.2

Findings:

- ▶ The three-knot model fits best, though barely better than one knot
- ▶ AIC values are **lower** than the corresponding proportional hazards models (Table from Fig 6.6 example) $\Rightarrow$ the **proportional odds model is preferred**

Treatment effect:

The treatment coefficient is 0.5287, so the odds ratio is $e^{0.5287} = 1.70$. The odds of surviving beyond any given time are **1.7 times higher** on tamoxifen.

This is fully consistent with the hazard ratio of 0.71 found under the Royston–Parmar model.

- ▶ **Flexible parametric models** bridge the Cox model and standard parametric models, giving a **smooth, parametric** baseline hazard and survival estimates at any time, including extrapolation.
- ▶ The **piecewise exponential model** holds the hazard constant between knots. With knots at every event time it matches the Cox model, but is best used with few (under 10) well-placed intervals, compared via AIC.
- ▶ **Spline functions** fit smoothly-joined polynomials between knots. **Cubic** splines balance flexibility and stability.
- ▶ **B-splines** are built from local basis functions via a recurrence relationship; **restricted (natural) cubic splines** add linearity beyond the boundary knots, so they are defined everywhere and need fewer basis functions.

- ▶ Knots are placed at percentiles of the uncensored event times; their number is chosen by **AIC or BIC**. Because models with different knot counts are **not nested**, likelihood ratio tests do not apply.
- ▶ Flexible **hazard** models build on $\log h_0(t)$ to keep the hazard positive, and the baseline is obtained by numerical integration.
- ▶ The **Royston–Parmar model** instead splines the **log-cumulative hazard** as a function of $\log t$, a Weibull model with extra knots. It still assumes proportional hazards, gives an **easily computed survivor function**, and yields a smooth baseline hazard. Variable selection and knot selection can be done separately.
- ▶ The same spline idea applied to the **log-odds** of survival gives **flexible proportional odds models**, which are particularly well suited to **unimodal** hazards.

Questions?

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